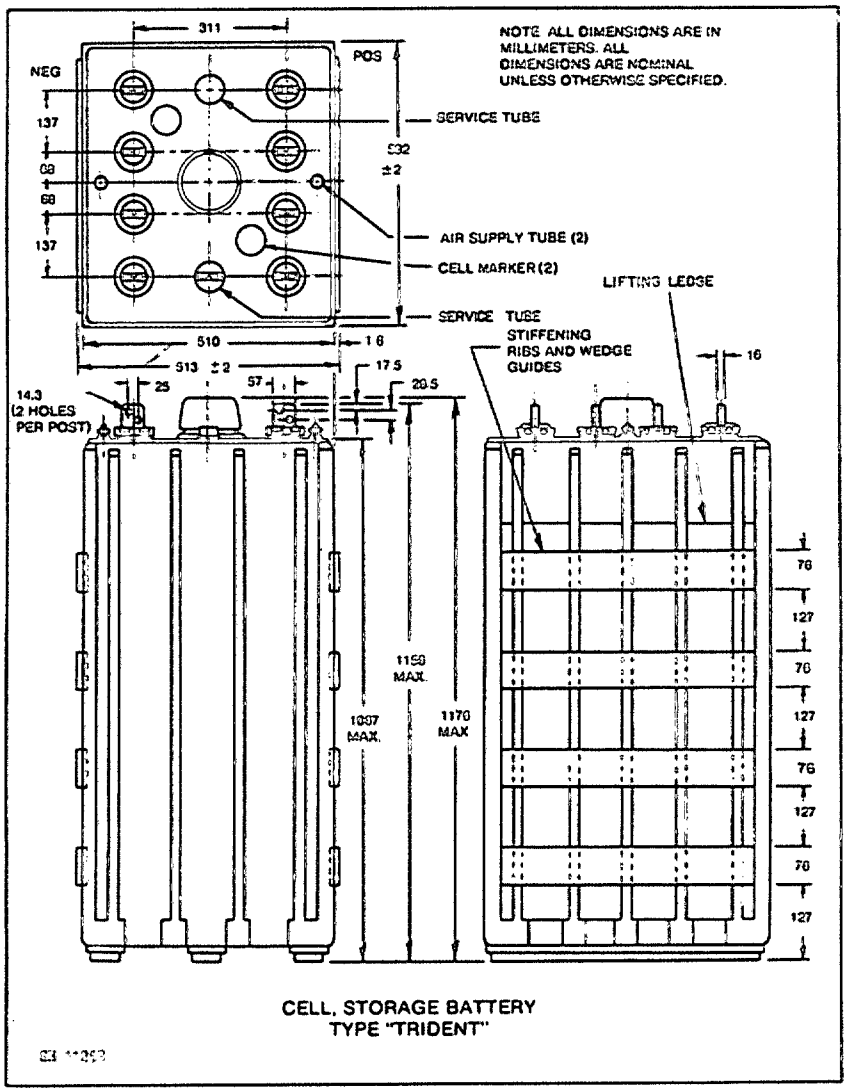
METRIC
DOD-B-24541/3A(SH)
24 November 1982
SUPERSEDING
DOD-B-24541/3(SH)
10 August 1976

MILITARY SPECIFICATION SHEET

BATTERY CELLS AND ELEMENTS, LEAD-ACID, MAIN STORAGE, SUBMARINE, TYPE TRIDENT (METRIC)

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery cells and elements described herein shall consist of this document and the latest issue of DOD-B-24541.



CELL, STORAGE BATTERY
TYPE "TRIDENT"

FSC 6140

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REQUIREMENTS:

Electrolyte specific gravity:	1.285 nominal.
Measurement of capacity:	Capacity, in ampere-hours (Ah), is based on an initial electrolyte temperature of 26.7 degrees Celsius, specific gravity of 1.285, and shall be not less than as specified in table I.
Water cooling:	Not required.

TABLE I. Electrical characteristics.

Capacity Ah	Time hours	Discharge current A	Minimum battery voltage		Minimum cell voltage	
			128-cell			
			Average	Final	Average	Final
7500	1	7500	211	183	1.46	1.28
10,350	3	3450	232	204	1.61	1.51
11,700	5	2340	240	210	1.65	1.57
12,150	6	2025	242	212	1.66	1.58
13,400	10	1340	245	215	1.69	1.63
14,600	20	720	248	217	1.71	1.66
15,300	36	425	250	220	1.72	1.68
15,600	48	325	251	220	1.72	1.68

Mass per cell (filled and charged):	Maximum - 1023 kilograms (kg) Maximum average - 1011 kg Minimum - 955 kg
Guaranteed service life: (equivalent cycle equals 10,000 Ah of charge)	200 equivalent cycles or 72 months, whichever occurs first.
Demonstration of rated capacity:	Within 50 equivalent cycles.
Life test:	Accelerated float charge and ship float simulation regimes.
Hydrogen emission (maximum): (measurements recorded as cubic centimeters/hour/cell/1,000 Ah at the 10-hour rate)	7,000 at 200 A 4,500 at 100 A 150 on open circuit
Trickle discharge rate:	10A

NOTE: Provide 2 electrolyte agitation systems per cell.

Preparing activity:
Navy - SH
(Project 6140-N563)

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